

ROSHAN DAHAL

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Links: [\[LinkedIn\]](#) | [\[Github\]](#) |

Highly motivated and detail-oriented Electronics, Communication, and Information Engineering graduate with a strong foundation in programming, database management, and networking principles. Possesses hands-on academic experience in developing complex hardware software integrated systems, machine learning models, and exploring distributed database architectures. Eager to leverage technical troubleshooting skills, knowledge of network protocols, and proficiency in Linux and database environments.

Bachelor in Electronics, Communication and Information Engineering

- Purwanchal Engineering Campus, Dharan, Nepal | (2078 BS-Present) *Completed 8th Semester Examinations*

Higher Secondary Education (10+2)

- Ambition Academy Secondary School | 2078BS (2021 AD) | GPA: 3.57

Secondary Education Examination (SEE)

- BB Memorial English Boarding School | 2075 BS (2019 AD) | GPA: 3.75

DOCUMENTS & VERIFICATION

- **Academic Certificates & Transcripts:**
[\[https://drive.google.com/file/d/1NwfQIDSSUpmMU_q_LDbWZQMY5J4jQ0DA/view?usp=sharing\]](https://drive.google.com/file/d/1NwfQIDSSUpmMU_q_LDbWZQMY5J4jQ0DA/view?usp=sharing)

TECHNICAL SKILLS

- **Programming & Scripting:** C, C++, JavaScript, Python, HTML, CSS
- **Frameworks & Libraries:** React, Node.js, Express.js, Next.js, TensorFlow/PyTorch
- **Databases:** PostgreSQL, MongoDB, SQL
- **Tools & Environments:** Linux, VS Code, Git
- **Languages:** Fluent in English and Nepali

ACADEMIC PROJECTS & PRACTICAL EXPERIENCE

Stock Price Prediction using LSTM

- Developed a Deep Learning model utilizing Long Short-Term Memory (LSTM) neural networks to forecast future stock prices based on historical financial market data.
- Conducted rigorous data preprocessing, feature scaling, and time-series analysis to train the model, effectively reducing predictive errors.
- Visualized predictive trends against actual market data to evaluate model accuracy and performance metrics.

Link: [<https://github.com/pawandai/stock-price-prediction.git>]

Autonomous Navigation Bot using Q-Learning

- Implemented a Reinforcement Learning algorithm (Q-Learning) allowing an autonomous robot to navigate dynamic environments and learn optimal obstacle-avoidance paths.
- Designed the reward-penalty system to train the bot in simulated environments, gradually improving its decision-making capabilities over successive epochs.
- Bridged complex algorithmic logic with sensor inputs for real-time environmental mapping and path optimization.

Link: [<https://github.com/BEI078/autonomous-navigation-bot.git>]

Data Networking & Distributed Systems Studies

- Researched and analyzed the technical principles of telecommunications and data networking, focusing on transmission technologies, signaling processes, and network latency.
- Studied distributed systems and large-scale data handling, including the architectural write flows of Hadoop Distributed File System (HDFS) and the hierarchical structure of Elasticsearch.

SOFT SKILLS

- Strong analytical and problem-solving abilities
- Effective technical documentation and presentation skills
- Adaptability and eagerness to learn new enterprise technologies
- Strong collaborative skills built through team-based engineering projects